

Czech ACM Student Chapter

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University of Žilina
Matej Bel University in Banská Bystrica

Czech Technical University in Prague

Technical University of Ostrava
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CTU Open Contest 2013

Fanatic Farmer Freddy

Freddy finds farming fullfilling. Not only that. He is literally obsessed with gardening. He spends all of his free time on his garden, which has become really huge and fabulous during the last years. While gardening, he often faces challenging problems. For some of them, a computer program would be extremely helpful. Will you help him to write such programs? Fresh cucumbers will be your reward!

Your programs can be written in C, C++, or Java programming languages. The choice is yours but you will be fully responsible for the correctness and efficiency of your solutions. We need the correct answer produced in some appropriate time. Nothing else matters. You may choose any algorithm and any programming style.

In the previous years, we found out that many teams do not read these introductory instructions. So we have a little bonus for those who do. We reveal here that the easiest problem is the first one. We recommend to solve it first.

All programs will read text from the standard input only. The results must be written to the standard output. You are not allowed to use any other files, communicate over network, or create processes. Input and output formats are described in problem statements and must be strictly followed. Do not print anything more than required. Each text line (including the last one) should be terminated by a newline character ("`\n`"), which is not considered to be a part of that line.

Good luck in the Czech Technical University Open Contest 2013!





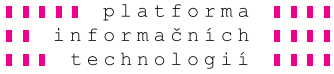












This problem set consists of nine sheets of paper (including this one) containing eight problems. Please make sure that you have the complete set.



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Fractional Lotion

`fl.c`, `fl.cpp`, `Fl.java`

Freddy practices various kinds of alternative medicine, such as homeopathy. This practice is based on the belief that successively diluting some substances in water or alcohol while shaking them thoroughly produces remedies for many diseases.

This year, Freddy's vegetables appear to have caught some disease and he decided to experiment a little bit and investigate whether homeopathy works for vegetables too. As Freddy is also a big fan of mathematics, he does not strictly insist that the substances have small concentrations, but he instead requires the concentrations to be reciprocals of integers ($1/n$). In experiments, some of the vegetables really got much better.

Seeing Freddy's successes, a fellow gardener also wants to try one of these potions and asks for a flask. Freddy has one flask of the potion in concentration $1/n$ and does not want to give it all out. Your task is to find out in how many ways the potion can be split into two flasks and diluted so that the resulting potions both have the same volume as the original one and the resulting concentrations also are reciprocals of integers — we do not want to end up with useless fluid, do we?

Input Specification

Each line of the input describes one test case. The line contains the expression "`1/`n" representing the original concentration. You are guaranteed that $1 \leq n \leq 10\,000$. There are no spaces on the line.

Output Specification

For each test case, output a single line with the total number of distinct pairs $\{x, y\}$ of positive integers satisfying $1/x + 1/y = 1/n$. Pairs differing only in the order of the two numbers are not considered different.

Sample Input	Output for Sample Input
`1/2`	`2`
`1/4`	`3`
`1/1`	`1`
`1/5000`	`32`

This problem was inspired by Project Euler

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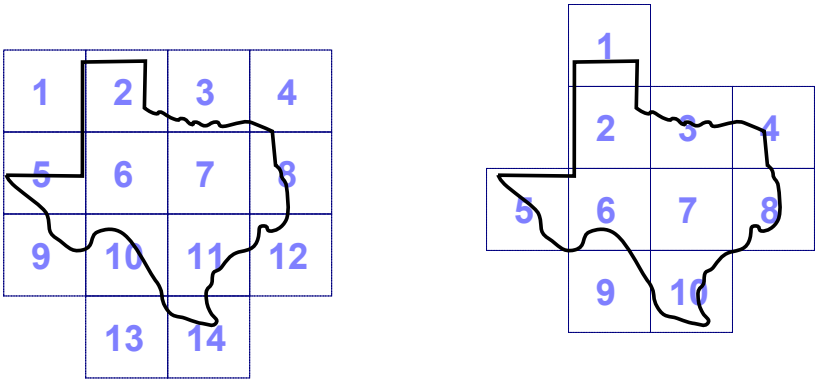
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Folded Map

`fm.c`, `fm.cpp`, `Fm.java`

Freddy's garden became so large that he needs a map to keep evidence of which vegetables are planted in what area. He ordered a high-quality map from the International Cartographic Publishing Company (ICPC). Since the map has a large scale, it does not fit onto a single page and it has to be split into several rectangular tiles.

Even with a fixed tile size (determined by a page size) and map scale, the number of tiles may still differ by adjusting the position of the tile grid. Your task is to find the minimal number of tiles necessary to cover the whole region of Freddy's garden.



Two of many possible ways of tiling Texas-shaped region

Let's have a look at an example. The figure on the left shows 14 map tiles covering a region. By adjusting the grid position a little bit, we may cover the same region with only 10 tiles, without changing their size or orientation.

Note that the tiles must be part of a rectangular grid aligned with the x-axis and y-axis. That is, they touch each other only with their whole sides and cannot be rotated.

Input Specification

The input contains several test cases. The first line of each test case contains four integer numbers: A_r, A_c, T_r, and T_c. A_r and A_c give the input image resolution in pixels ($1 \leq A_x \leq 1000$), while T_r and T_c is the size of one tile in pixels ($1 \leq T_x \leq 100$). The next A_r lines each contain A_c characters, each of them being either "`X`" (the pixel corresponds to a part of the garden to be covered by a tile) or "`.`" (the corresponding pixel is outside the garden and does not need to be covered). The region pixels form one *connected* region.

Output Specification

For each test case, print one integer number — the minimal number of tiles necessary to cover all pixels represented by "X".

Sample Input

```
3 3 2 2
XXX
XXX
XXX
3 3 2 2
XX.
XXX
XXX
17 32 5 9
........XXXXXXXX................
........XXXXXXXX................
........XXXXXXXX................
........XXXXXXXXX...............
........XXXXXXXXXXXXXXX.........
........XXXXXXXXXXXXXXXXXXX....
........XXXXXXXXXXXXXXXXXXXX...
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX..
..XXXXXXXXXXXXXXXXXXXXXXXXXXXXX.
....XXXXXXXXXXXXXXXXXXXXXXXXXXX.
......XXXXXXXXXXXXXXXXXXXXXXXXX.
........XX..XXXXXXXXXXXXXXXXX...
..............XXXXXXXXXXXXXX.....
................XXXXXXXXX........
..................XXXXXXX........
....................XXXXX........
......................XXX........
```

Output for Sample Input

```
4
3
10
```



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Furry Nuisance

`fn.c`, `fn.cpp`, `Fn.java`

In order to protect himself from evil bunnies, Freddy decided to install an automatic system to detect them in pictures from surveillance cameras. Sophisticated software detects important points in the picture and lines between them. Unfortunately, the terrain in the pictures is quite varied and lot of the points and lines are actually not bunnies.

You have made the following observation: Each bunny has four paws and a body joining them. Based on this observation, write a program to decide whether a given picture can possibly contain a bunny.

Input Specification

The input contains several test cases. The first line of each test case contains two integers n and m ($0 \leq n \leq 10\,000$, $0 \leq m \leq 20\,000$), giving the number of points and lines in the image, respectively. Each of the m following lines contains two distinct integers x and y ($1 \leq x, y \leq n$), indicating that the points x and y are directly joined by a line. You may assume that each pair of points is joined by at most one direct line and that no point is directly joined with itself.

Output Specification

For each input instance, output "`YES`" if the picture can contain a bunny, and "`NO`" otherwise. The picture can contain a bunny if it is possible to remove some of the points and lines so that the resulting image is connected and has exactly 4 paws.

The image is said to be connected if (and only if) each two points are joined with each other by one or more successive lines. A paw is a point which is directly joined with exactly one other point.

Sample Input

```
2 1
1 2
5 4
1 2
1 3
1 4
1 5
```

Output for Sample Input

```
NO
YES
```



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Fence Orthogonality

`fo.c`, `fo.cpp`, `Fo.java`

Evil bunnies are eating Freddy's vegetables. In order to stop them, he decided to build a fence enclosing all vegetables in his garden. Freddy wants the fence to be as cheap (i.e., short) as possible, but for technical reasons, he can only build rectangular fences. For simplicity, we will assume the vegetables are negligibly small and can be represented by points in a two-dimensional plane.

Input Specification

The input consists of several test cases. The first line of each test case contains one integer N ($3 \leq N \leq 10\,000$) giving the number of vegetables in the garden. Each of the following N lines contains two integers X_i and Y_i ($0 \leq X_i, Y_i \leq 10\,000$), giving the coordinates of one vegetable to be protected. No two vegetables have the same coordinates. You may also assume the vegetables are not *all* on the same straight line.

Output Specification

For each test case, output a single line containing one real number t, giving the smallest length of the perimeter of a rectangular fence enclosing all the vegetables. Note that the edges of the rectangle do not need to be parallel with the coordinate axes.

The answer will be accepted as correct if the difference between t and the exact answer is at most 0.0005.

Sample Input

```
3
0 0
1 0
0 1
3
10 0
0 10
4 4
4
1 0
0 1
2 1
1 2
```

Output for Sample Input

```
4
31.112698
5.656854
```

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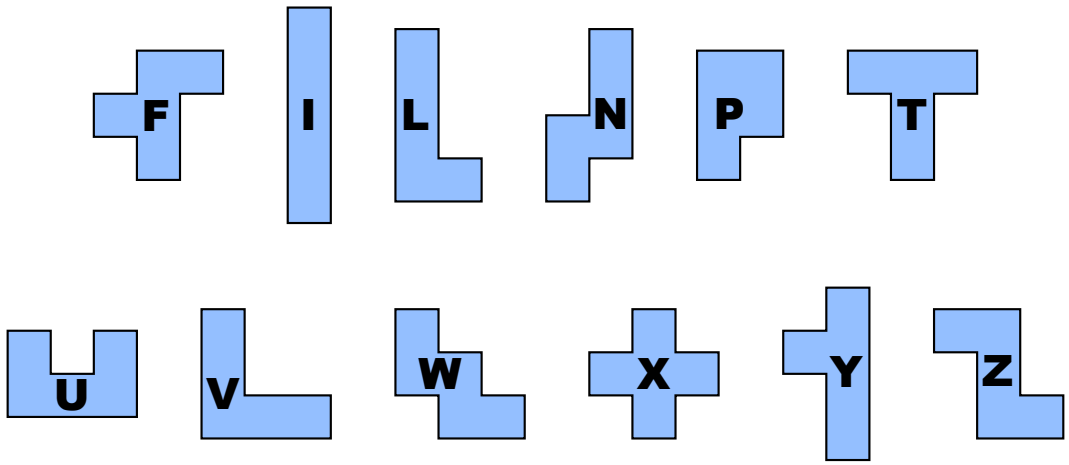
Flower Pots

`fp.c`, `fp.cpp`, `Fp.java`

Freddy is planning a garden party for this weekend. Many old classmates are going to pay him a visit and Freddy is thinking about some elegant and unexpected improvement of his small horticultural oasis. Inspired by the creations in his favorite gardening journals he decided to install two flower arrangements on the opposite sides of the walkway leading to his garden. To match the other plants in the vicinity, one arrangement has to be in light bronze yellow and the other in dark pastel red color. The pots in which the flowers will be placed have to be painted with the same matching colors.

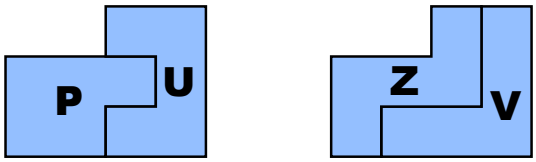
Freddy knows that it is much cheaper and faster to buy new pots than to repaint existing ones. Thus, being pressed by fast-approaching date of the party, Freddy has decided to purchase flower pots from a small company which not only can deliver pots painted with exact colors needed but they can deliver them today afternoon. The owner of the company is a Dutch artist who specializes herself in designs far from ordinary. Her very special pots are manufactured in so-called pentomino shapes and they can be flipped over so that their top side can serve as the bottom side and vice versa.

A pentomino-shaped flower pot is made of five squares welded together so that sides of any two neighboring squares always touch each other along the whole edge. In the actual pots, each square is about ten square feet, but the size does not matter for this problem, it is only important that all squares in all shapes are of the same size. There are exactly 12 possible shapes, they are listed in the figure below and each shape is traditionally named by a letter to which it bears some resemblance. (with Freddy's favorite shape being **F**, of course)



Freddy is going to buy two yellow pots and two red pots. For aesthetical reasons, he wants both arrangements to be of equal shape. Two pots of the same color will be put on the lawn closely together so that the divisions between them will not be visible and only the outline of the whole arrangement will be important for judging the equality of shapes. In the resulting arrangements, no pots can overlap.

Freddy has to choose the pots carefully because some pairs are clearly incompatible from this point of view (such as W+F and I+I), some other may be compatible but it might not be immediately obvious that they really are (such as P+U and V+Z and their possible arrangement in the figure below). Therefore, Freddy asks you for a help. You are given two yellow pots and two red pots and your program should decide if two arrangements with the same outline can be created.



Input Specification

The input contains several test cases. Each test case consists of one line. The line starts with two letters that specify the shapes of two yellow pots, then there is one space and other two letters giving the shapes of two red pots. All four letters are in uppercase and each of them is one of the 12 valid letters listed above.

Output Specification

For each test case, output a single line of text. The line should contain "`YES`" if an arrangement exists which can be composed from both pairs of yellow pots and red pots. If there is no such arrangement, the line should contain "`NO`".

Sample Input

```
II PP
WF II
VZ UP
```

Output for Sample Input

```
YES
NO
YES
```



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Frustrated Queue

`fq.c`, `fq.cpp`, `Fq.java`

The toilet in Freddy's garden is broken, so his only chance are public toilets nearby. One day, there is a long queue of people in front of the toilets. Freddy is in a big need and so he desperately wants the queue to be served as quick as possible.

To use the toilets, you need to pay 5 crowns. Half of the people in the queue have a 5-crown coin and the other half only have a 10-crown coin. Initially, the toilet operator has no coins, thus, the people in the queue have to reorganize so that whenever someone wants to pay with a 10-crown coin, the operator has at least one 5-crown coin available from previous customers.

The issue is that some of the people in the queue are unwilling to give up their spot. Determine in how many ways can the people willing to change their position rearrange themselves in the queue so that the operator always has change available. The positions of those unwilling cannot change (they cannot be moved to a later but also to an *earlier* spot in the rearranged queue). Furthermore, among those willing to change the position, the relative order of those with the same coin must be preserved.

Input Specification

The input contains several test cases. Each test case consists of one line containing a non-empty string of parentheses and dots of length $n \leq 1\,000$. A dot indicates a person willing to change their position in the queue, an opening parenthesis indicates a person unwilling to change the position who has a 5-crown coin, and a closing parenthesis indicates a person unwilling to change the position who has a 10-crown coin.

You may assume that n is even and that the string contains at most $n/2$ opening parentheses and at most $n/2$ closing ones.

Output Specification

For each test case, compute the number of ways the queue can be rearranged so that the conditions described in the statement of the problem are satisfied. Since this number may be too large, you are only required to print a single line containing one integer equal to the last 6 digits (in the decimal representation) of the number.

Sample Input	Output for Sample Input
`....`	`2`
`.(..`	`1`
`)...`	`0`
`.....).....................`	`68484`



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Frozen Rose-Heads

`fr.c`, `fr.cpp`, `Fr.java`

The winter is coming and all the experts are warning that it will be the coldest one in the last hundred years. Freddy needs to make sure that his garden does not sustain any damage. One of the most important tasks is to make sure that no water remains in his large watering system.

All the water comes from a central node and is distributed by pipes to neighboring nodes and so on. Each node is either a sprinkler (rose head) with no outgoing pipe or an internal node with one or more outgoing pipes leading to some other nodes. Every node has exactly one incoming pipe, except for the central node which takes the water directly from a well and has no incoming pipe. Every pipe has a valve that stops all the water going through the pipe. The valves are of different quality and age, so some may be harder to close than others.

Freddy knows his valves well and has assigned a value to each pipe representing the amount of effort needed to close the corresponding valve. He asks you to help him count the minimum effort needed to close some valves so that no water goes to the sprinklers.

Input Specification

The input contains several test cases. Each test case starts with a line with two integers, the number of nodes n ($2 \leq n \leq 1\,000$), and the number of the central node c ($1 \leq c \leq n$). Each of the next $n-1$ lines represents one pipe and contains three integers, u, v ($1 \leq u, v \leq n$) and w ($1 \leq w \leq 1\,000$), where u and v are the nodes connected by a pipe and w is the effort needed to close the valve on that pipe. You may assume that every node is reachable from the central node.

Output Specification

For each test case, output a single line containing the minimum sum of efforts of valves to be closed, such that the central node gets separated from all sprinklers.

Sample Input

```
3 1
2 1 5
1 3 4
7 7
7 6 10
7 5 10
6 4 1
6 3 1
5 2 1
5 1 2
```

Output for Sample Input

```
9
5
```

This problem was adapted from Stanford Local Contest

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False Sense of Security

`fs.c`, `fs.cpp`, `Fs.java`

Freddy discovered a new procedure to grow much bigger cauliflowers. He wants to share this finding with his fellow gardener Tommy but he does not want anyone to steal the procedure. So the two gardeners agreed upon using a simple encryption technique proposed by M. E. Ohaver.

The encryption is based on the Morse code, which represents characters as variable-length sequences of dots and dashes. The following table shows the Morse code sequences for all letters:

A	`.-`	H	`....`	O	`---`	V	`...-`
B	`-...`	I	`..`	P	`.--.`	W	`.--`
C	`-.-.`	J	`.---`	Q	`--.-`	X	`-..-`
D	`-..`	K	`-.-`	R	`.-.`	Y	`-.--`
E	`.`	L	`.-..`	S	`...`	Z	`--..`
F	`..-.`	M	`--`	T	`-`		
G	`--.`	N	`-.`	U	`..-`		

Note that four possible dot-dash combinations are unassigned. For the purposes of this problem we will assign them as follows (note these are not the assignments for actual Morse code):

underscore (“_”)	`..--`	period (“.”)	`---.`
comma (“,”)	`.-.-`	question mark (“?”)	`----`

In practice, characters in a message are delimited by short pauses, typically displayed as spaces. Thus, the message ACM_GREATER_NY_REGION is encoded as:
`.- -.-. -- ..-- --. .-. . .- - . .-. ..-- -. -.-- ..-- .-. . --. .. --- -.`

The Ohaver's encryption scheme is based on mutilating Morse code, namely by removing the pauses between letters. Since the pauses are necessary (because Morse is a variable-length encoding that is not prefix-free), a string is added that identifies the number of dots and dashes in each character. For example, consider the message “`.--.-.--`”. Without knowing where the pauses should be, this could be “`ACM`”, “`ANK`”, or several other possibilities. If we add length information, such as “`.--.-.-- 242`”, then the code is unambiguous.

Ohaver's scheme has three steps, the same for encryption and decryption:

1. Convert the text to Morse code without pauses but with a string of numbers to indicate code lengths.
2. Reverse the string of numbers.
3. Convert the dots and dashes back into the text using the reversed string of numbers as code lengths.

As an example, consider the encrypted message "AKADTOF_IBOETATUK_IJN". Converting to Morse code with a length string yields:
.--.-.--..----..-...--..-...---.-.--..--.-..--...----. 232313442431121334242
By reversing the numbers and decoding, we get the original message "ACM_GREATER_NY_REGION".

The security of this encoding scheme is not too high but Freddy believes it is sufficient for his purposes. Will you help Freddy to implement this encoding algorithm and to protect his sensitive information?

Input Specification

The input will consist of several messages encoded with Ohaver's algorithm, each of them on one line. Each message will use only the twenty-six capital letters, underscores, commas, periods, and question marks. Messages will not exceed 1000 characters in length.

Output Specification

For each message in the input, output the decoded message on one line.

Sample Input

```
FENDSVTSLHW.EDATS,EULAY
TRDNWPLOEF
NTTTGAZEJUIIGDUZEHKUE
QEWOISE.EIVCAEFNRXTBELYTGD.
?EJHUT.TSMYGW?EJHOT
DSU.XFNCJEVE.OE_UJDXNO_YHU?VIDWDHPDJIKXZT?E
ADAWEKHZN,OTEATWRZMZN_IDWCZGTEPION
```

Output for Sample Input

```
FALSE_SENSE_OF_SECURITY
CTU_PRAGUE
TWO_THOUSAND_THIRTEEN
QUOTH_THE_RAVEN,_NEVERMORE.
TO_BE_OR_NOT_TO_BE?
THE_QUICK_BROWN_FOX_JUMPS_OVER_THE_LAZY_DOG
ADAPTED_FROM_ACM_GREATER_NY_REGION
```